

Owen Chambers

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EDUCATION

University of Waterloo

Waterloo, Ontario, Canada

MMath in Computer Science - 3.7/4

2021–2023

- Thesis: User-specific explanations of AI systems attuned to psychological profiles: a user study.

Brandon University

Brandon, Manitoba, Canada

Honors B.Sc Computer Science - 3.83/4.3

2016–2021

- Thesis: Algorithmic Composition: Generative AI for Classical Music Synthesis.

PUBLICATIONS

- [1] O. Chambers, R. Cohen, M. R. Grossman, L. Hebert, and E. Awad, “Mining user study data to judge the merit of a model for supporting user-specific explanations of ai systems”, *Computational Intelligence*, vol. 40, no. 6, e70015, 2024.
- [2] O. Chambers, “User-specific explanations of ai systems attuned to psychological profiles: A user study”, 2023.
- [3] O. Chambers, R. Cohen, M. R. Grossman, and Q. Chen, “Creating a user model to support user-specific explanations of ai systems”, in *Adjunct Proceedings of the 30th ACM Conference on User Modeling, Adaptation and Personalization*, 2022, pp. 163–166.

RESEARCH

User-specific Explanations of AI Systems

University of Waterloo Thesis

- Designed and conducted a behavioral user study ($N = 150$) psychometrically profiling participants to model human acceptance/rejection thresholds of AI-generated explanations
- Trained and tested a PyTorch-based Feedforward Neural Network on the study results. This AI model yielded **statistically significant** results when predicting the acceptance/rejection of AI explanations
- Applied **Linear Regression, Logistic Regression, and Random Forest AI models** to gain further insight into how the models predicted whether a participant would accept a given explanation.

Predicting Off-target CRISPR Mutations

Deep-Learning for Biotechnology Course Project

- Developed a **Siamese Neural Network** using Pytorch to predict off-target CRISPR mutations achieving **98% accuracy** improving on previous research
- Introduced a novel dual-stream architecture that independently processes gRNA and DNA sequence strands through parallel neural networks, optimizing weight updates to significantly outperform traditional combined-dataset baselines

Tailoring Product Recommendations for Mobile Devices

Information Retrieval Course Project

- Created a React Native application capable of scraping Amazon and displaying product images and descriptions

- This project investigated the relationship between image size and text volume on a persons product preference on mobile devices. A user study was conducted with 20 participants and statistical analysis was performed on the results

Generative AI in Music Composition

Brandon University Thesis

- Extracted high-dimensional feature representations from classical piano compositions, utilizing vector embeddings to map variations in theme, tempo, cadence, and stylistic attributes
- Developed a novel algorithm to arrange and improvise on these pieces to create new music based on a persons preferences for speed, and consistency

EXPERIENCE

Software Developer

BrainRidge Consulting Inc

Nov 2024–Present

- Developed an agentic AI orchestrator capable of spawning processes and sub-processes for a given requirement until the task was completed
- This covered the entire software development life cycle including creating requirements from specs, integrating with MCP servers, writing, testing, and iterating on code, and performing AWS infrastructure changes.

Machine Learning Contractor

Invisible Technologies

May 2024–Oct 2024

- Fine-tuned state-of-the-art LLMs and agentic architectures for autonomous web navigation and software development, achieving a 20% benchmark accuracy improvement.
- Evaluated model performance, managed dataset curation/alignment, and engineered complex reward functions/prompts in collaboration with frontier AI research labs (OpenAI, Microsoft).

Software Developer Intern

Technology Solve

Oct 2020–Jan 2021

- Engineered relational data models and optimized search/retrieval algorithms for an inventory tracking application, minimizing database query latency
- Analyzed web traffic patterns and executed data-driven SEO optimizations for enterprise clients (100+ employees), yielding a 12% increase in user engagement

SKILLS

- **Language Familiarity:** Typescript, Python, JavaScript, React/React Native, SQL, HTML, CSS, Scripting
- **Technologies:** PyTorch, Scikit-learn, Tensorflow, Numpy, Pandas, AWS